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## MA 651: Optimization Techniques

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| Course Code   | MA 651                      |
| Course Name   | Optimization Techniques     |
| L-T-P-C       | 3-0-0-3                     |
| Prerequisites | None                        |
| Intended for  | Ph.D./ M.S./B.Tech          |
| Distribution  | Elective                    |
| Approval      | 2nd Adhoc Senate (May 2012) |

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### Course Contents

- **Module I** [6 Lectures]  
Convex sets and function, Introduction to optimization, Model formulation, Simplex based techniques, Concept of duality.
- **Module II** [10 lectures]  
Quadratic Programming Problem, Geometric Programming, Separable Programming.
- **Module III** [12 lectures]  
Direct and Gradient based search techniques for single and multi variable unconstrained optimization problems.
- **Module IV** [6 lectures]  
Penalty and barrier function based techniques for constrained optimization problems.
- **Module V** [6 lectures]  
Evolutionary Optimization Techniques, Engineering application of Optimization techniques.

### Text and Reference Books

1. Mokhtar S. Bazaraa, Hanif D. Sherali and M.C.Shetty, **Nonlinear Programming, Theory and Algorithms**, John Wiley & Sons, 2004.
2. S. S. Rao, **Engineering Optimization: Theory and Practice**, 4th Edition, John Wiley & Sons, 2009.
3. Kwang Y. Lee, Mohamed A. El-Sharkawi, **Modern heuristic optimization techniques: theory and applications**, Kluwer, 2008.
4. Hamdy A. Taha, **Operations Research: An Introduction**, 8th Edition, Pearson Education, 2008.
5. G. V. Reklaitis, A. Ravindran, K. M. Ragsdell, **Engineering Optimization: Methods and Applications**, Wiley, 2006.
6. Michael C. Bartholomew-Biggs, **Nonlinear optimization with engineering applications**, Springer, 2008.